

A Proof of Conjecture 00000000117: Pandigital Primes in Every Base from 2 to 16

Feiyu*

Claude (Opus 5.5)[†]

2026-10-02

Abstract

Conjecture 00000000117 asserts that for every base $b \in [2, 16]$ there is a b -ary pandigital prime, a prime whose base- b expansion contains each digit $0, \dots, b-1$ at least once. We exhibit one in every base: the smallest one, as listed in OEIS A185122. We verify the digits of each witness, and we prove its primality with Pratt certificates checked against the Lucas primality criterion. Everything is formalized in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000000117 reads:

Conjecture: For every base $b \in [2, 16]$ there exists a b -ary pandigital prime (whose expansion contains each of the digits $0, \dots, b-1$ at least once).

2 Reading of the statement

For an integer $b \geq 2$, the base- b expansion of a positive integer n is its usual digit string without leading zeros. n is *b -ary pandigital* if every $d \in \{0, \dots, b-1\}$ occurs in it. The conjecture asserts that for each integer b with $2 \leq b \leq 16$ some prime is b -ary pandigital. The statement is finite and unambiguous.

3 Result

Theorem 1. *For each $b \in \{2, \dots, 16\}$ the number p_b of Table 1 is a b -ary pandigital prime. Hence Conjecture 00000000117 holds.*

4 Proof

Digits. Reading off the expansions in Table 1, each contains every digit $0, \dots, b-1$. (In Lean, p_b is written as $\sum_i d_i b^i$ for the listed digit string, every digit is checked to be $< b$ with a nonzero leading digit, so the string is *the* base- b expansion of p_b , and membership of each digit is then checked directly.)

*Independent researcher.

[†]Anthropic.

b	p_b	base- b expansion of p_b
2	2	10
3	11	102
4	283	10123
5	3319	101234
6	48761	1013425
7	863231	10223465
8	17119607	101234567
9	393474749	1012346785
10	10123457689	10123457689
11	290522736467	1022345689A7
12	8989787252711	101234568A79B
13	304978405943587	10123456789ABC
14	11177758345241723	10123456789CDAB
15	442074237951168419	10223456789ADBCE
16	18528729602926047181	10123456789ABEFCD

Table 1: The witnesses: the smallest b -ary pandigital primes (OEIS A185122). Digits 10, ..., 15 are written A–F.

Primality. The witnesses $p_2, \dots, p_7$ are below 10^6 and their primality is checked by trial division. For the others we use the classical criterion of Lucas, in the form proved in Mathlib.

Lemma 2 (Lucas). *Let $p \geq 2$ and a be integers with $a^{p-1} \equiv 1 \pmod{p}$ and $a^{(p-1)/q} \not\equiv 1 \pmod{p}$ for every prime $q \mid p-1$. Then p is prime.*

Proof. The order of a in $(\mathbb{Z}/p\mathbb{Z})^\times$ divides $p-1$ but no $(p-1)/q$, so it equals $p-1$. Hence $(\mathbb{Z}/p\mathbb{Z})^\times$ has $p-1$ elements and every nonzero residue is a unit, so p is prime. $\square$

To apply Lemma 2 one needs all prime factors of $p-1$, which in turn are certified prime in the same way unless they are small: this is a Pratt certificate [1]. Table 2 lists every certificate used. For each row, $p-1$ equals the displayed product, the witness a satisfies the two congruences of Lemma 2, and each factor is either below 10^6 (prime by trial division) or itself a row of the table.

The modular powers $a^e \bmod p$ with p up to $1.85 \cdot 10^{19}$ are computed by binary exponentiation, so each check takes about 65 multiplications modulo p .

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) states the conjecture and proves it against Mathlib.

- `IsPandigital b p` — every $d < b$ occurs in `Nat.digits b p`, Mathlib’s base- b expansion.
- `ConjectureHolds` — every b with $2 \leq b \leq 16$ has a prime p with `IsPandigital b p`.
- `powMod` — binary exponentiation modulo m , with `powMod_eq`: it equals $a^e \bmod m$ whenever $e < 2^{\text{fuel}}$.
- `mem_of_prime_dvd_prod` — a prime dividing $\prod q_i^{e_i}$ (with all q_i prime) is one of the q_i .
- `prime_of_cert` — Lemma 2 with a factored $p-1$, derived from Mathlib’s `lucas_primality`.

p	a	factorization of $p - 1$
1065901	6	$2^2 \cdot 3 \cdot 5^2 \cdot 11 \cdot 17 \cdot 19$
1222829	2	$2^2 \cdot 347 \cdot 881$
6395407	3	$2 \cdot 3 \cdot 1065901$
17119607	5	$2 \cdot 7 \cdot 1222829$
67222271	7	$2 \cdot 5 \cdot 2087 \cdot 3221$
335476601	6	$2^3 \cdot 5^2 \cdot 47 \cdot 89 \cdot 401$
393474749	2	$2^2 \cdot 691 \cdot 142357$
10123457689	13	$2^3 \cdot 3^2 \cdot 4091 \cdot 34369$
290522736467	2	$2 \cdot 433 \cdot 335476601$
8989787252711	11	$2 \cdot 5 \cdot 11 \cdot 131617 \cdot 620933$
35485169204621	2	$2^2 \cdot 5 \cdot 41 \cdot 197 \cdot 317 \cdot 347 \cdot 1997$
50829734323931	2	$2 \cdot 5 \cdot 1129 \cdot 13873 \cdot 324529$
132708276772139	2	$2 \cdot 19 \cdot 47 \cdot 170179 \cdot 436627$
304978405943587	2	$2 \cdot 3 \cdot 50829734323931$
11177758345241723	2	$2 \cdot 13 \cdot 6395407 \cdot 67222271$
442074237951168419	2	$2 \cdot 6229 \cdot 35485169204621$
18528729602926047181	6	$2^2 \cdot 3 \cdot 5 \cdot 13 \cdot 179 \cdot 132708276772139$

Table 2: Pratt certificates, in increasing order of p .

- `prime_1065901, ..., prime_18528729602926047181` — the rows of Table 2.
- `digits_b, isPandigital_b` for $b = 2, \dots, 16$ — the expansions of Table 1, via Mathlib’s `Nat.digits_ofDigits`.
- `conjecture_00000000117` — `ConjectureHolds`.

All numerical checks are carried out by the Lean kernel, with `decide +kernel` and `norm_num`. The kernel evaluates numeral arithmetic natively, and `powMod` keeps every intermediate value below p^2 . The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds in a few seconds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000000117
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Remark 3 (why the witnesses repeat a digit). For $b \geq 4$, no pandigital prime has exactly b digits. If each digit occurs once, then $n \equiv 0 + 1 + \dots + (b - 1) = b(b - 1)/2 \pmod{b - 1}$. For even b this is $\equiv 0$, so $b - 1 \mid n$. For odd b it is $\equiv (b - 1)/2$, so $(b - 1)/2 \mid n$. In both cases n has a proper divisor greater than 1 once $b \geq 4$. This is why every witness in Table 1 with $b \geq 4$ has $b + 1$ digits.

Remark 4. The witnesses are the smallest pandigital primes, as recorded in OEIS A185122 [2]; a computer search confirms this, but minimality is not needed and is not part of the Lean development. Any pandigital prime in each base would do.

References

- [1] V. R. Pratt, *Every prime has a succinct certificate*, SIAM J. Comput. 4 (1975), 214–220.
- [2] OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, sequence A185122 (minimum pandigital prime in base n).